

Word Sense Disambiguation

- The task of selecting the correct sense for a word
- **Input**
 - ✓ a word in context
 - ✓ a fixed inventory of potential word senses
- **Output**
 - ✓ the correct word sense in context

All-words Tasks

- Need to disambiguate all the words in some text
 - the system is given an entire text
 - a lexicon with an inventory of senses for each entry

✓ similar to **part-of-speech tagging**, except with a much larger set of tags
✓ each lemma has its own set.

Supervised All-words Tasks

- Generally trained from a **semantic concordance**
 - ✓ a corpus
 - ✓ each open-class word in each sentence is labeled with its word sense from a specific dictionary or thesaurus, most often WordNet.

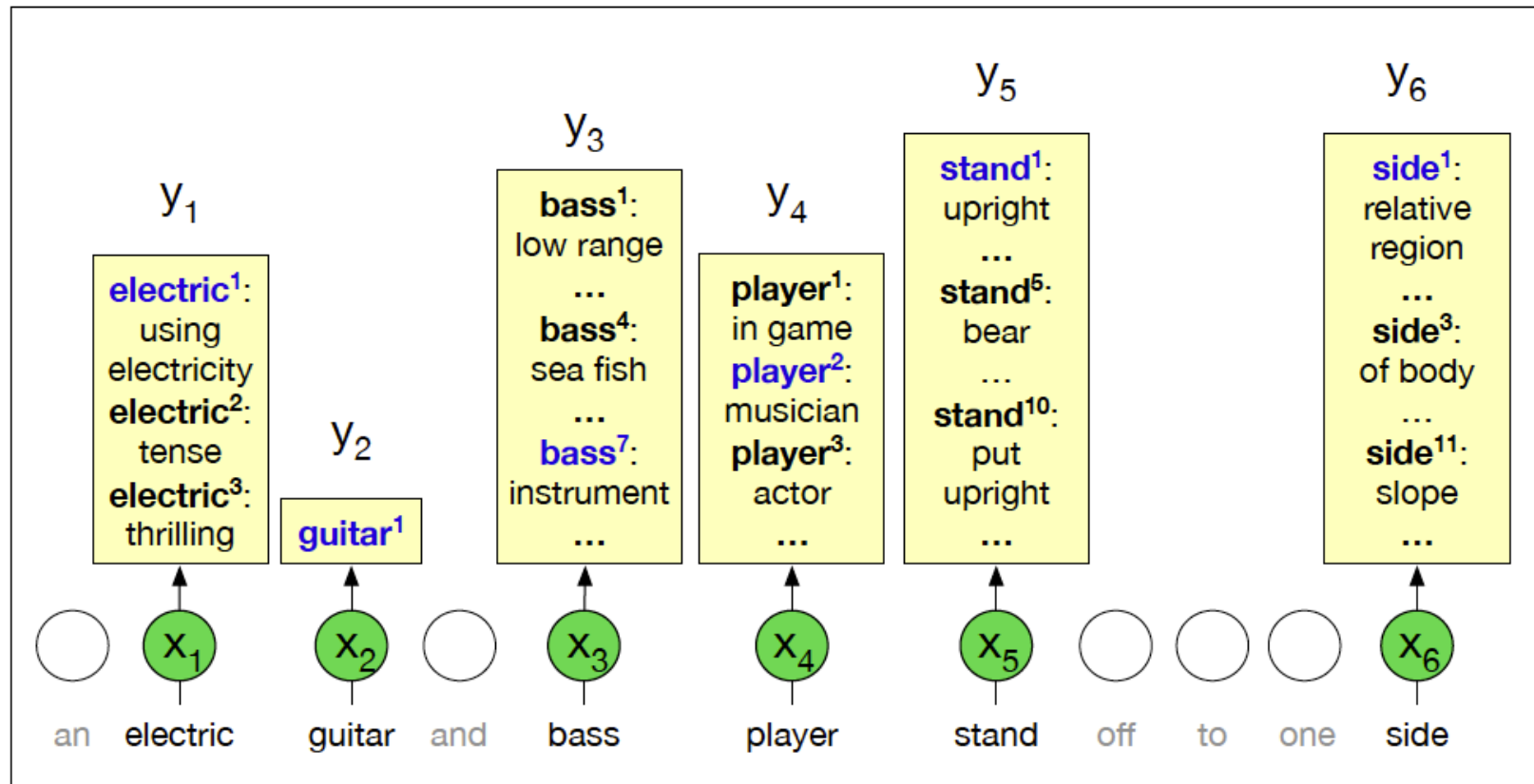
Supervised All-words Tasks

- SemCor corpus
- SENSEVAL-3 Task 1 corpus
- SemEval-13 Task 12 corpus

You will find_v⁹ that avocado_n¹ is_v¹ unlike_j¹ other_j¹ fruit_n¹ you have ever_r¹ tasted_v²

Given each noun, verb, adjective, or adverb word in the hand-labeled test set (say fruit), the SemCor-based WSD task is to choose the correct sense from the possible senses in WordNet

Supervised All-words Tasks



All-words Tasks - **Baseline**

- Simply choose the **most frequent sense** for each word from the senses in a labeled corpus.
- For WordNet, this corresponds to the **first sense**
- Senses in WordNet are generally ordered from most frequent to least frequent based on their counts.

The WSD Algorithm: - **Lesk Algorithm**

- Knowledge-based algorithms
- Rely solely on WordNet or other such resources
- Don't require labeled data, like SemCor

*choose the sense whose dictionary **gloss or example** shares the most words with the target word's neighborhood.*

Lesk Algorithm

The *bank* can guarantee deposits will eventually cover future tuition costs because it invests in adjustable-rate mortgage

bank ¹	Gloss:	a financial institution that accepts deposits and channels the money into lending activities
	Examples:	“he cashed a check at the bank”, “that bank holds the mortgage on my home”
bank ²	Gloss:	sloping land (especially the slope beside a body of water)
	Examples:	“they pulled the canoe up on the bank”, “he sat on the bank of the river and watched the currents”

WordNet Senses

Lesk Algorithm

function SIMPLIFIED LESK(*word*, *sentence*) **returns** best sense of *word*

best-sense $\leftarrow$ most frequent sense for *word*

max-overlap $\leftarrow$ 0

context $\leftarrow$ set of words in *sentence*

for each *sense* **in** senses of *word* **do**

signature $\leftarrow$ set of words in the gloss and examples of *sense*

overlap $\leftarrow$ COMPUTEOVERLAP(*signature*, *context*)

if *overlap* > *max-overlap* **then**

max-overlap $\leftarrow$ *overlap*

best-sense $\leftarrow$ *sense*

end

return(*best-sense*)

Lesk Algorithm

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WordNet Senses

Lesk Algorithm

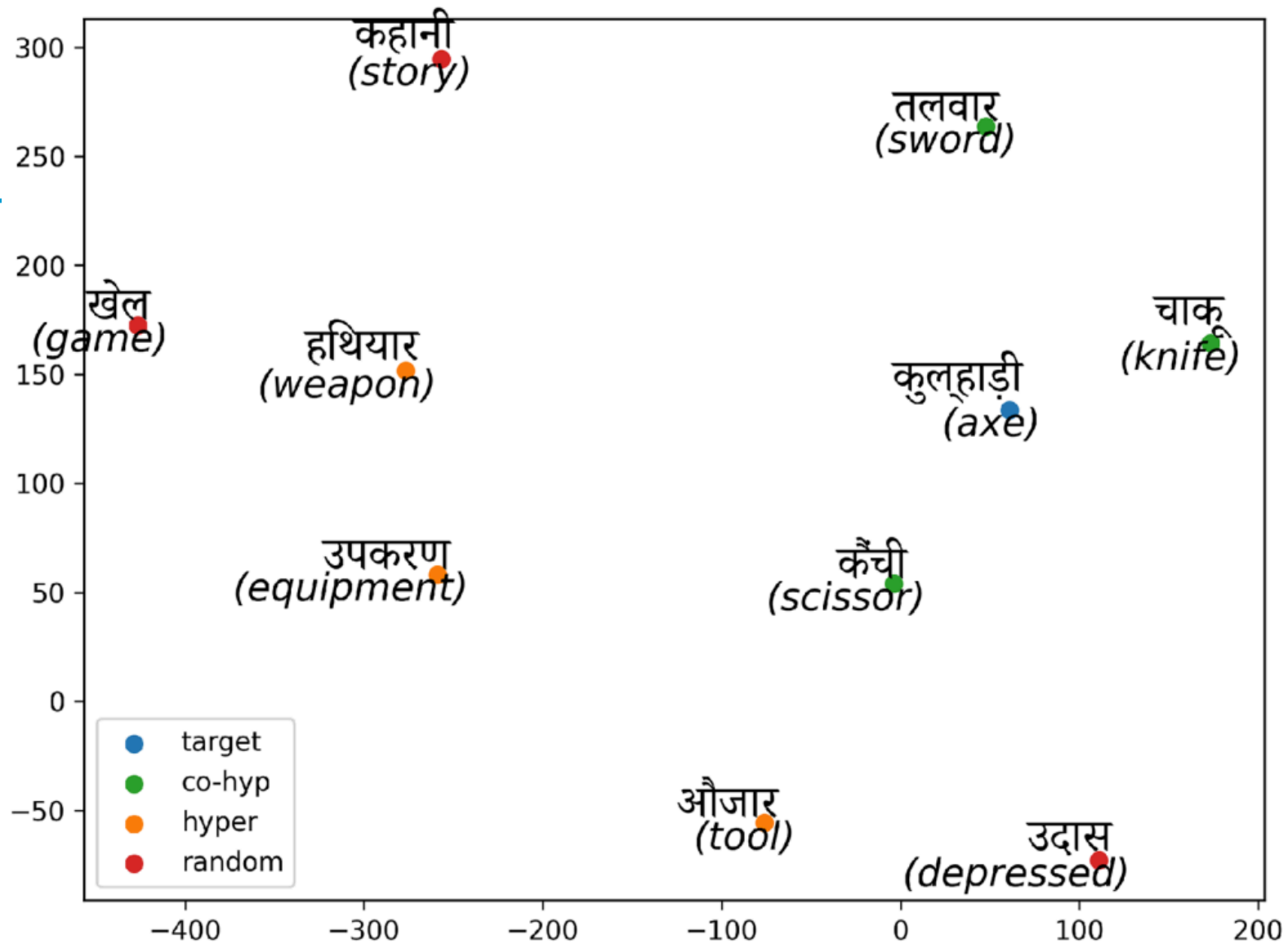
The *bank* can guarantee *deposits* will eventually cover future tuition costs because it invests in adjustable-rate *mortgage*

bank ¹	Gloss:	a financial institution that accepts <i>deposits</i> and channels the money into lending activities
	Examples:	“he cashed a check at the bank”, “that bank holds the <i>mortgage</i> on my home”
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WordNet Senses

Lesk Algorithm

What can be improved further?



The WSD Algorithm: **Feature-Based WSD**

- **IMS algorithm** (Zhong and Ng, 2010)
- Classification Task, SVM Classifier (**Self Reading**)
- Uses simple features of the surrounding words
 - ✓ **part-of-speech tags** (for a window of 3 words on each side, stopping at sentence boundaries)
 - ✓ **collocation features** of words or n-grams of lengths 1, 2, 3 at a particular location in a window of 3 words on each side
 - ✓ weighted average of **embeddings**

The WSD Algorithm: **Feature-Based WSD**

An electric guitar and **bass** player stand off to one side

$$[w_{i-2}, \text{POS}_{i-2}, w_{i-1}, \text{POS}_{i-1}, w_{i+1}, \text{POS}_{i+1}, w_{i+2}, \text{POS}_{i+2}, w_{i-2}^{i-1}, \\ w_{i+1}^{i+2}, g(E(w_{i-2}), E(w_{i-1}), E(w_{i+1}), E(w_{i+2}))]$$

[guitar, NN, and, CC, player, NN, stand, VB, guitar and,
player stand, g(E(guitar), E(and), E(player), E(stand))]

The WSD Algorithm: **Contextual Embeddings**

- 1-nearest-neighbor algorithm using contextual word embeddings.
- **Steps**
 - ✓ At training time we pass each sentence in the SemCore labeled dataset through any contextual embedding producing model
 - ✓ Resulting in a contextual embedding for each labeled token in SemCore.

The WSD Algorithm: **Contextual Embeddings**

- **Steps**

- ✓ For each sense ***s*** of any word in the corpus, for each of the ***n*** tokens of that sense, we average their *n* contextual representations ***v_i*** to produce a contextual sense embedding ***v_s*** for ***s***:

$$\mathbf{v}_s = \frac{1}{n} \sum_i \mathbf{v}_i \quad \forall \mathbf{v}_i \in \text{tokens}(s)$$

The WSD Algorithm: **Contextual Embeddings**

- **Steps**

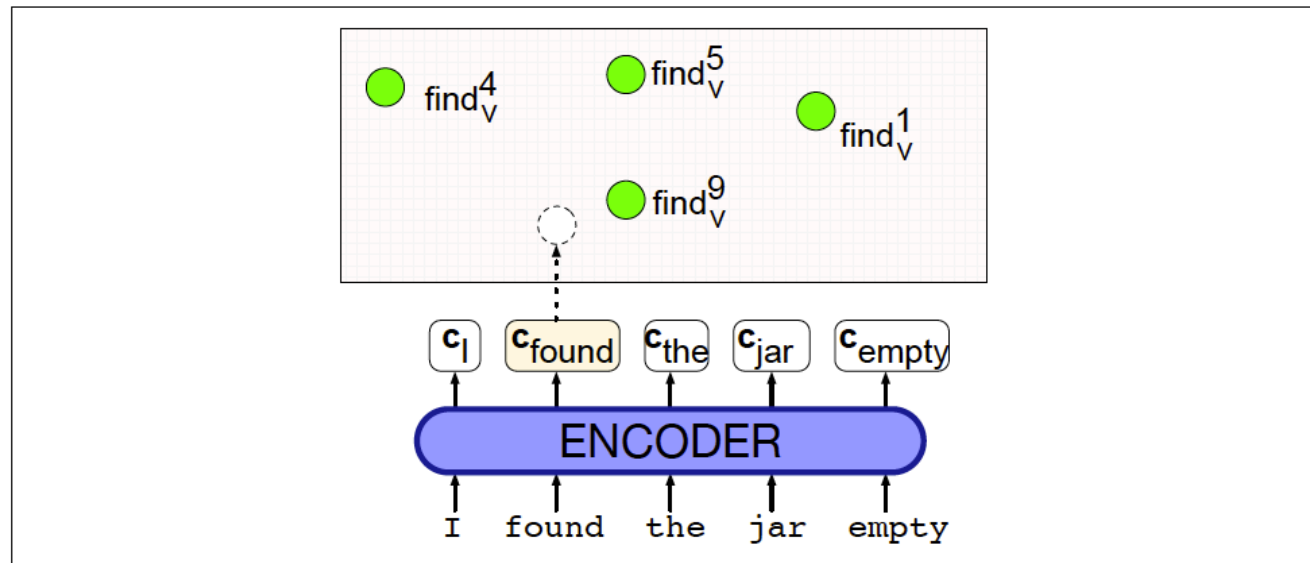
- ✓ At test time, given a token of a target word t in context, compute its contextual embedding $\mathbf{t}$
- ✓ choose its nearest neighbour sense from the training set

$$\text{sense}(t) = \underset{s \in \text{senses}(t)}{\operatorname{argmax}} \cos(\mathbf{t}, \mathbf{v}_s)$$

The WSD Algorithm: **Contextual Embeddings**

- **Steps**

- ✓ At test time, given a token of a target word t in context, compute its contextual embedding t
- ✓ choose its nearest neighbour sense from the training set



Word-in-context task

- Input
 - ✓ Two sentences, each with the same target word but in a different sentential context.
- Output
 - ✓ Decide whether the target words are used in the same sense in the two sentences or in a different sense.

Word-in-context task : Dataset

- WiC dataset (Pilehvar and Camacho-Collados, 2019).

F There's a lot of trash on the **bed** of the river —

I keep a glass of water next to my **bed** when I sleep

F **Justify** the margins — The end **justifies** the means

T **Air** pollution — Open a window and let in some **air**

T The expanded **window** will give us time to catch the thieves —

You have a two-hour **window** of clear weather to finish working on the lawn

Word-in-context task

- compute the contextual embeddings for the target word in each of the two sentences
- compute the cosine similarity between them.
- If it's above a threshold tuned on a devset
 - respond **true** (the two senses are the same)
 - else we respond **false**.

Other source of training data

- Wikipedia as a source of sense-labeled data
- When a concept is mentioned in a Wikipedia article,
 - ✓ the article text may contain an explicit link to the concept's Wikipedia page, which is named by a unique identifier.
 - ✓ This link can be used as a sense annotation.

Other source of training data

In 1834, Sumner was admitted to the **[[bar (law)|bar]]** at the age of twenty-three, and entered private practice in Boston.

It is danced in 3/4 time (like most waltzes), with the couple turning approx. 180 degrees every **[[bar (music)|bar]]**.

Jenga is a popular beer in the **[[bar (establishment)|bar]]s** of Thailand.

Other source of training data

- Collect and add these sentences to the training data for a supervised system
- Map from Wikipedia concepts to whatever inventory of senses.
 - ✓ finding the WordNet sense that has the greatest lexical overlap with the Wikipedia sense
 - ✓ by comparing the vector of words in the WordNet synset, gloss, and related senses with the vector of words in the Wikipedia page title, outgoing links, and page category ([Ponzetto and Navigli, 2010](#)).
 - ✓ The resulting mapping has been used to create **BabelNet**, a large sense-annotated resource ([Navigli and Ponzetto, 2012](#)).

Vector Semantics

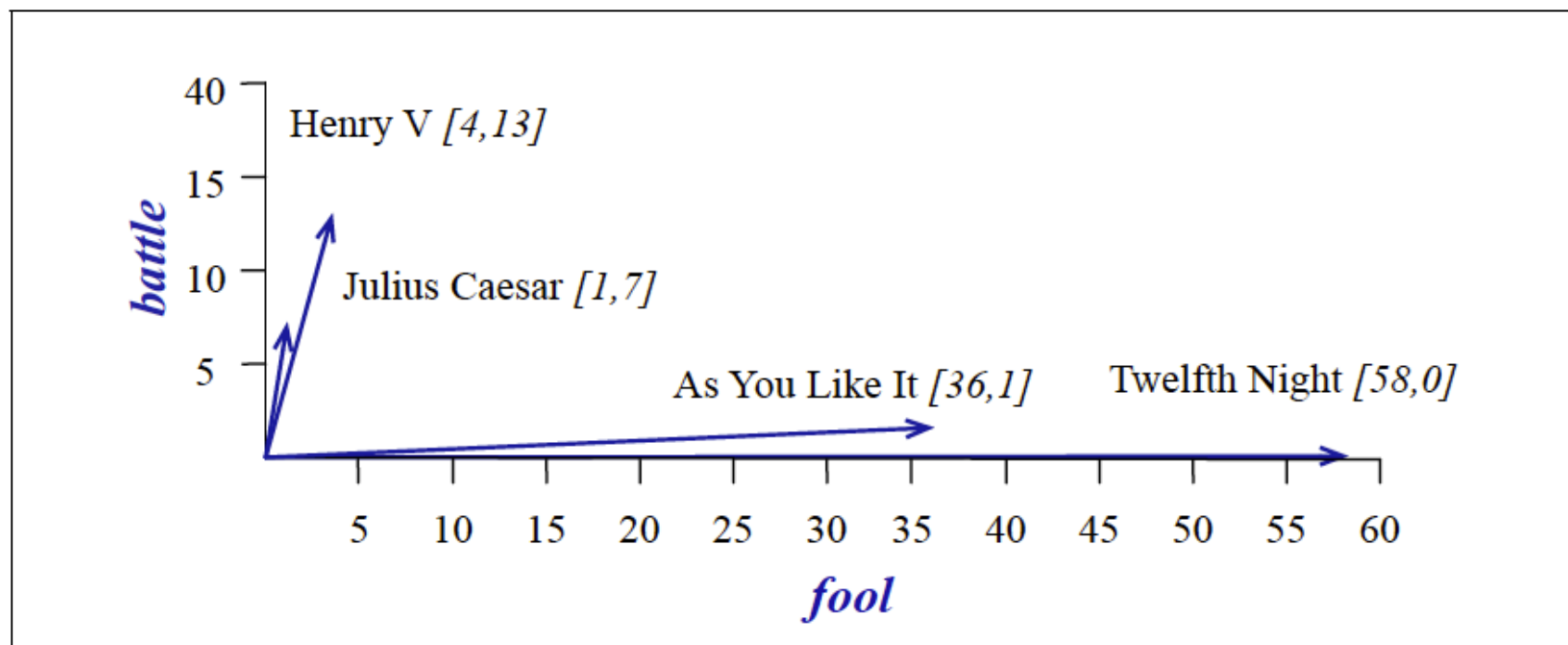
- Vector or distributional models of meaning are generally based on a co-occurrence matrix
 - ✓ term-document matrix
 - ✓ term-term matrix

Term-Document Matrix

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

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Term-Document Matrix



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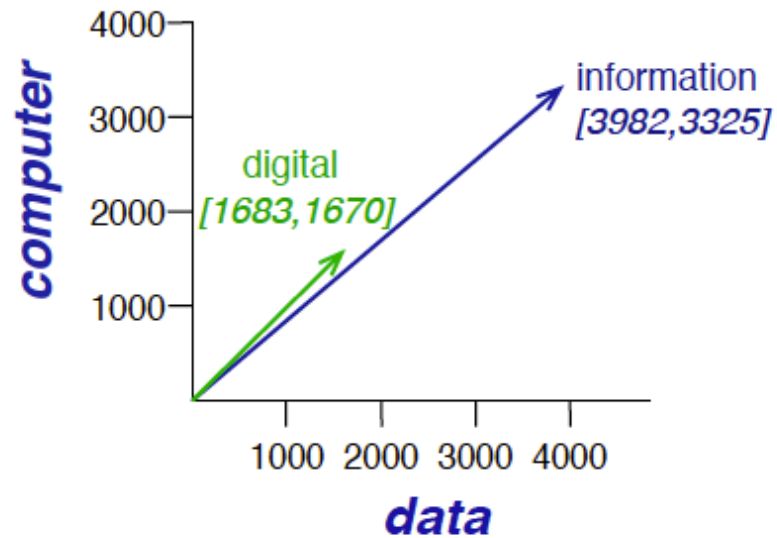
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Term-Term Matrix

is traditionally followed by **cherry** pie, a traditional dessert
often mixed, such as **strawberry** rhubarb pie. Apple pie
computer peripherals and personal **digital** assistants. These devices usually
a computer. This includes **information** available on the internet

	aardvark	...	computer	data	result	pie	sugar	...
cherry	0	...	2	8	9	442	25	...
strawberry	0	...	0	0	1	60	19	...
digital	0	...	1670	1683	85	5	4	...
<u>information</u>	0	...	3325	3982	378	5	13	...

Term-Term Matrix



Term-Term Matrix

	pie	data	computer
cherry	442	8	2
digital	5	1683	1670
information	5	3982	3325

cos (cherry, information)= ??
cos (digital, information)= ??

$$\cos(\text{cherry}, \text{information}) = \frac{442 * 5 + 8 * 3982 + 2 * 3325}{\sqrt{442^2 + 8^2 + 2^2} \sqrt{5^2 + 3982^2 + 3325^2}} = .018$$

$$\cos(\text{digital}, \text{information}) = \frac{5 * 5 + 1683 * 3982 + 1670 * 3325}{\sqrt{5^2 + 1683^2 + 1670^2} \sqrt{5^2 + 3982^2 + 3325^2}} = .996$$

Term-Term Matrix

